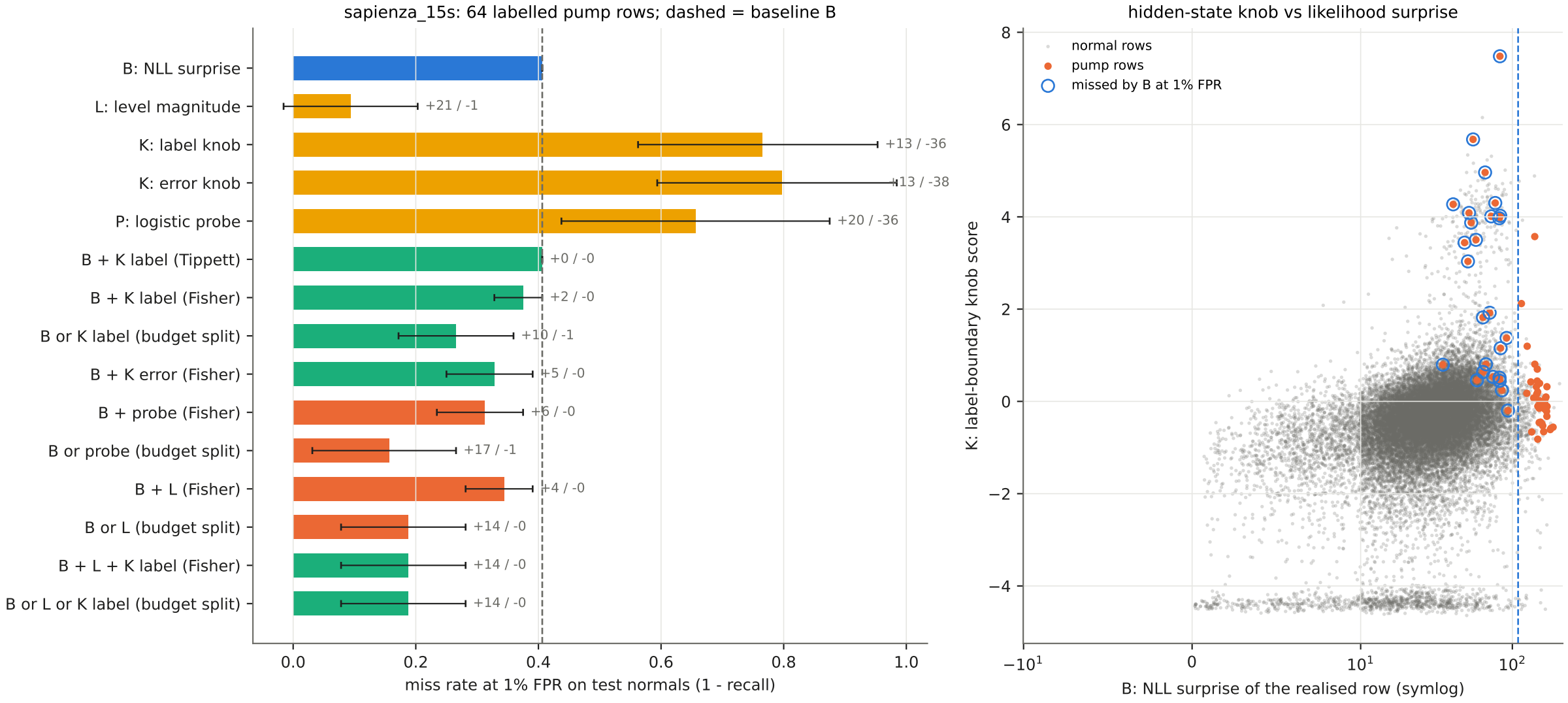

sapienza_15s: 64 labelled pump rows; dashed = baseline B
B: NLL surprise
L: level magnitude
K: label knob
K: error knob
P: logistic probe
B + K label (Tippett)
B + K label (Fisher)
B or K label (budget split)
B + K error (Fisher)
B + probe (Fisher)
B or probe (budget split)
B + L (Fisher)
B or L (budget split)
B + L + K label (Fisher)
B or L or K label (budget split)
+21 / -1
+13 / -36
+13 / -38
+20 / -36
+0 / -0
+2 / -0
+10 / -1
+5 / -0
+6 / -0
+17 / -1
+4 / -0
+14 / -0
+14 / -0
+14 / -0
0.0
0.2
0.4
0.6
0.8
1.0
miss rate at 1% FPR on test normals (1 - recall)
hidden-state knob vs likelihood surprise
normal rows
pump rows
missed by B at 1% FPR
K: label-boundary knob score
8
6
4
2
0
−2
−4
−10^1
0
10^1
10^2
B: NLL surprise of the realised row (symlog)